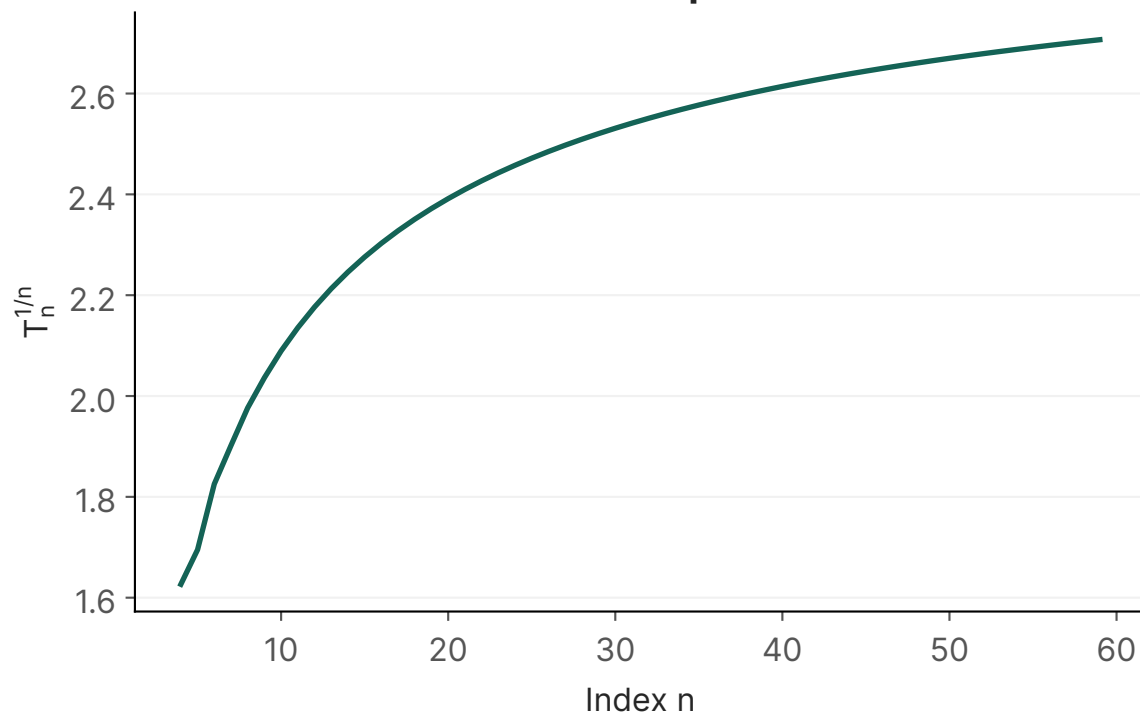


Alternating Motzkin roots: a sharp threshold

$r_n = T_n^{1/n}$, with margin $\Delta_n = 2\log r_n - \log r_{n-1} - \log r_{n+1}$

The root sequence



Lean proves every center $n \geq 6$ is positive and center 5 is negative.

The plot shows $n = 4 \dots 59$; heights are numerical, with every displayed sign checked by exact integer powers.

Log-concavity margin (symmetric log scale)

